Sahaj Bhadja

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Summary

New-grad software engineer with 2+ years building **full-stack product systems**, **large-scale data workflows**, and **quality-focused AI tooling** using Python, TypeScript, React, and cloud-native infrastructure.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Built **data-driven backend services** and interfaces for 500+ users, reducing latency from **1.2s to 350ms** through SQL optimization and API performance tuning.
- Implemented **quality-guardrail orchestration** for 5K+ monthly workflows, improving reliability and reducing production defects across rapidly evolving product features.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Developed **full-stack React + Flask features** handling 10K+ daily requests, converting product hypotheses into shippable APIs and UI flows through fast iteration.
- Improved **CI/CD and release quality loops**, reducing turnaround by **60%** via automated checks, structured triage, and proactive stakeholder communication.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Built **large-scale evaluation pipelines** over 100K+ trajectories, enabling reproducible quality estimation and ranking-style analyses for model behavior diagnostics.
- Produced **technical summaries and failure analyses** that improved collaboration between research and engineering teams and accelerated action on key findings.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Engineered **distributed API and ETL services** with Python/Java and AWS tools, improving response times by **30%** across high-volume operational systems.
- Maintained **20+ automated test suites** using Jenkins, PyTest, and Mockito, improving pipeline efficiency by **68%** and reducing regression risk.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Built **MERN + WebSocket modules** supporting eight workflows and 2M+ synchronized records, improving task throughput and real-time user responsiveness.
- Automated **containerized deployment workflows** with Docker/Jenkins, reducing release time by **50%** and improving deployment consistency.

Skills

Languages: **Python**, **TypeScript/JavaScript**, Java, SQL, C++, Bash
Product Engineering: full-stack web apps, API integration, feature iteration, customer/stakeholder feedback loops
Data & AI Systems: large-scale task processing, quality estimation workflows, RAG/LLM tooling fundamentals
Infra: AWS, Kubernetes basics, MongoDB/MySQL, Docker, GitHub Actions, CI/CD
Engineering: code reviews, debugging large codebases, extensible design, clear written/verbal communication

Projects

Labeler Quality Prediction Pipeline | *Python, XGBoost, PostgreSQL, FastAPI*
- Built a **quality prediction workflow** to estimate annotator output quality and prioritize review queues, improving throughput and confidence for high-volume labeling tasks.

Customer Support RAG Assistant | *Python, LangChain, Vector DB, React*
- Developed a **RAG-powered support application** handling thousands of queries with citation-backed answers, reducing manual support load and improving response consistency.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*
- Shipped a **scalable full-stack platform** for 50K+ users and 1M+ transactions, emphasizing resilient APIs, production observability, and rapid product iteration.